Junior Makers · Week 3

Mini Weather Station

Measure the weather, show it on a screen!

Ages 8-11 | Arduino + 2 sensors + OLED | Data science



Did you know?
Weather stations all over the world share readings every hour so forecasters can predict tomorrow.

Let's think...

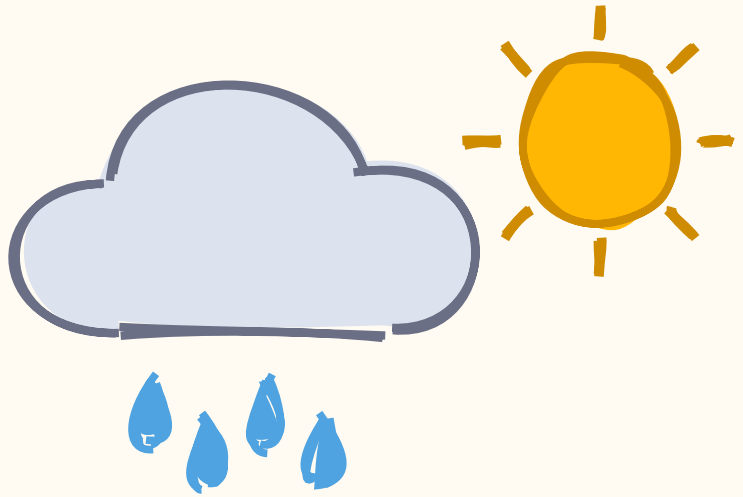
Cricket match tomorrow. Will it rain?

You are the match organiser

It is hot and sticky today. Dark clouds are coming.

Do you cancel the match, or play?

What information would help you decide?



How hot is it?

How wet is the air?

Is it sunny or dark?

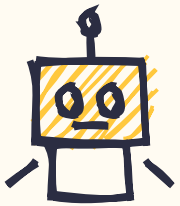
Today: we build a machine that measures all three!

Learning aims

Today I will learn to...

1. Explain **temperature**, **humidity** and **light**
2. Connect a **DHT22** and a **light sensor** to one Arduino
3. Show three readings on an **OLED** screen
4. Record readings in a **table** and compare them
5. Explain **why** readings change in different places

Keywords
temperature · °C · humidity · % · sensor · data · display



Watch and think

Video time!



Weather Tools for Kids

Give Spark · about 3 minutes · youtu.be/oQB68rvRWBM

Watch for...

1. Which tool measures **temperature**?
2. Which tool measures **wind**?
3. Why do farmers and pilots need weather data?

Scan to watch

After the video: Which of these tools will our mini weather station copy? Tell your partner.

The science

What does a weather station measure?



Temperature

How hot or cold the air is

unit: °C



Humidity

How much water vapour is in the air

unit: %



Light

How bright the sky is

unit: level

Big weather stations also measure

wind speed · wind direction · rainfall · air pressure

In real life

The Pakistan Meteorological Department runs weather stations across the country and issues heatwave and rain warnings.

Real world

Who needs weather data?

Farmers

When to water and harvest crops

Pilots

Safe take-off and landing

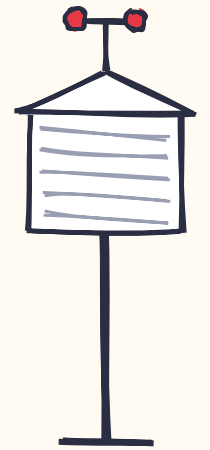
Doctors

Heatwave warnings keep people safe

Families

What to wear, umbrella or not

? Where does the weather app on a phone get its numbers? From **thousands of weather stations** just like the one we are building!



Measure 1

Temperature: how hot is the air?



Temperature	What it feels like
0 °C	Ice melts
25 °C	A nice spring day
37 °C	Your body
40 °C+	Islamabad in June, often!
100 °C	Water boils

How the sensor knows

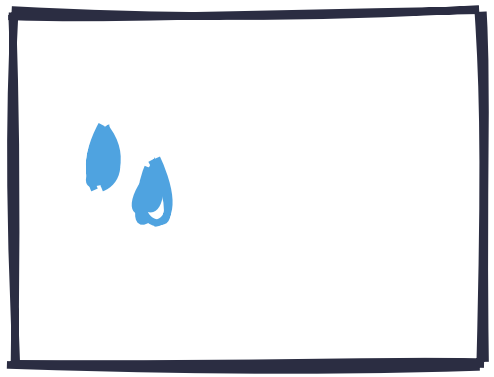
Inside the DHT22 is a tiny part whose **resistance changes with heat**. The chip turns that change into a number in °C.

Try it

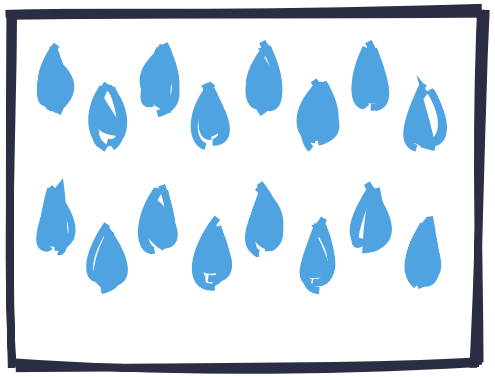
Hold the sensor gently in your hand. Watch the temperature climb towards your skin temperature!

Measure 2

Humidity: how wet is the air?



Dry air: 20%



Humid air: 85%

Air carries invisible **water vapour**. Humidity tells us how full of water the air is, from **0%** (very dry) to **100%** (cannot hold any more: fog or rain!).

Feel it

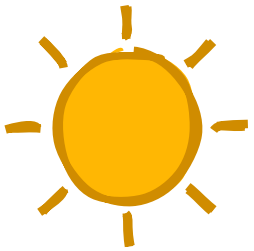
High humidity makes a hot day feel **sticky**, because your sweat cannot dry.
Breathe on a mirror: that mist is water vapour!

In real life

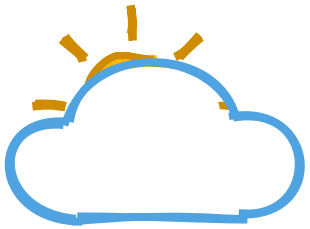
During the monsoon, humidity in Pakistan often goes above 80%. Farmers watch it closely.

Measure 3

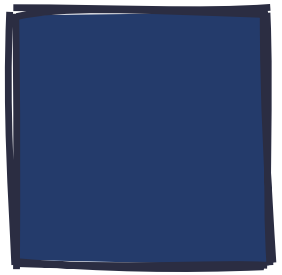
Light: sunny, cloudy or dark?



Sunny: high



Cloudy: medium



Dark: low

Why measure light?

Solar panels make more power in bright light.

Plants need light to grow.

Dark clouds can mean rain is coming.



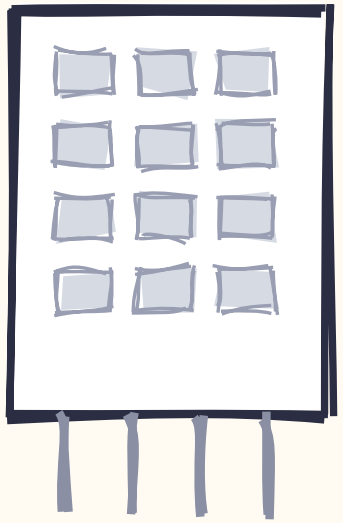
Our **LDR light sensor** sends the Arduino a number from **0 to 1023**. We call it the **light level**.

Heads up

Some light modules give a lower number in bright light. Test yours first!

Meet the sensor

The DHT22: two sensors in one



pins, left to right:

VCC · DATA · NC · GND

Pin	Connect to
VCC	5V power
DATA	Pin 2 (sends the readings)
NC	Not connected
GND	Ground

Some DHT22 boards have only 3 pins and a built-in resistor. If yours has 4 pins, add a 10 kΩ resistor between VCC and DATA.

Fact file

Temperature: **-40 to 80 °C**

Humidity: **0 to 100%**

New reading: every **2 seconds**

Why wait 2 seconds?

The DHT22 is slow. Asking faster gives old or empty readings!

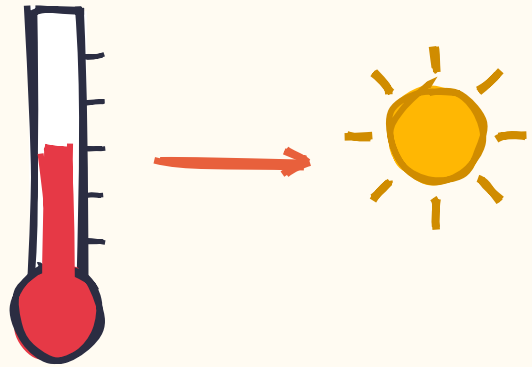
Be a data scientist

Take readings in three places

Place	Temp (°C)	Humidity (%)	Light
Classroom			
Outside in the sun			
Outside in the shade			
My guess: hottest place?			

Predict first
Scientists guess BEFORE they measure, then check. Write your guess in the last row!

Wait **2 minutes** in each place before you write the reading. Sensors need time to settle!



Explain the data

Why are the readings different?



Sun: Direct sunshine heats the sensor and the ground around it.



Shade and trees: Trees block the sun and release water, so shade is cooler and more humid.



Indoors: Walls, fans and people change the air inside our classroom.

Sentence starter

"The _____ was hottest because _____."

In real life

Cities plant trees to cool hot streets. Green spaces help fight heatwaves (SDG 13: Climate Action).

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Our weather station kit (per pair)
2. Wiring the weather station
3. The weather station in ArduinoBlocks
4. Make the screen easy to read
5. Challenges for weather scientists

Build it

Our weather station kit (per pair)



Arduino Uno
The brain



DHT22
Temperature + humidity



LDR module
Light level



OLED screen
Our weather display



Wires + breadboard
Roads for data



Data table
Pencil and paper!

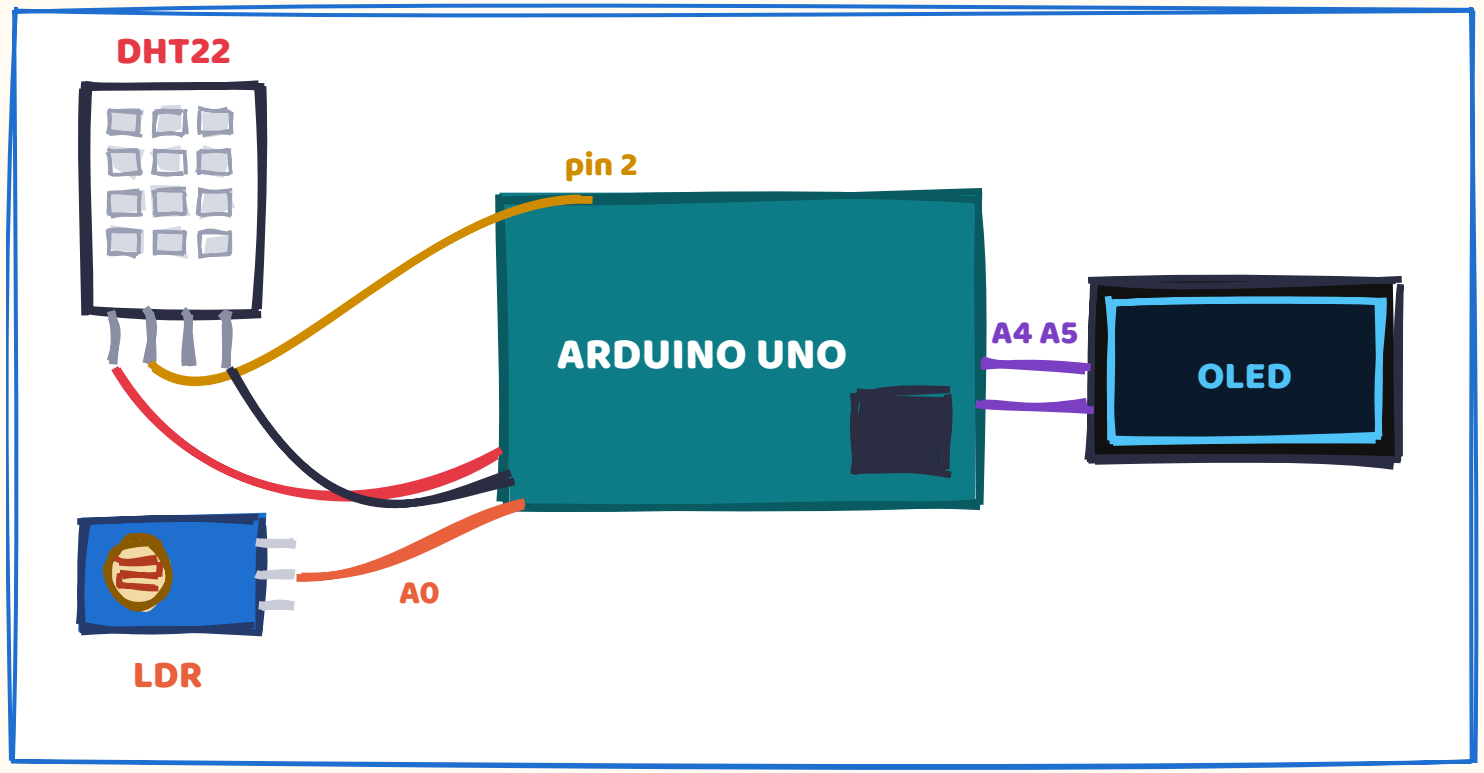
Plus: 10 kΩ resistor (for a 4-pin DHT22), jumper wires, USB cable, laptop.

In real life
Real weather stations keep sensors in a white box with slatted sides, so the sun does not heat them directly.

MAKE

Build it

Wiring the weather station



Part	Arduino
DHT22 VCC / DATA / GND	5V / pin 2 / GND
LDR VCC / A0 / GND	5V / A0 / GND
OLED VCC / GND	5V / GND
OLED SDA / SCL	A4 / A5

Test first

Build it in Wokwi: it has a DHT22 and OLED you can test before touching real parts.

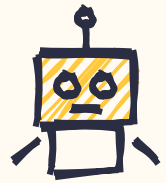
Code it

The weather station in ArduinoBlocks

```
when Arduino starts
    set up OLED screen
forever
    set temp to DHT22 temperature (pin 2)
    set hum to DHT22 humidity (pin 2)
    set light to read analog A0
    clear OLED
    OLED print 'Temp: ' + temp + ' C'
    OLED print 'Hum: ' + hum + ' %'
    OLED print 'Light: ' + light
    if temp > 30 → show SUN icon
    wait 2000 ms
```

Read, then show

1. **Read** all three sensors into variables.
2. **Clear** the screen.
3. **Show** each reading with its unit.
4. **Wait** 2 seconds and repeat.



Common bug

Screen flickers or writes on top of itself? You forgot to clear the OLED before printing!

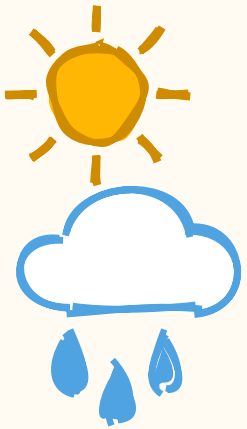
Design for the reader

Make the screen easy to read

```
MY WEATHER
Temp: 31.4 C (sun)
Hum: 62 %
Light: 740
Feels: HOT!
```

Good screen rules

- [x] Always show the **unit** (°C, %)
- [x] One reading per line
- [x] Use a **title** so people know what it is
- [x] Add a simple message or icon



In real life

Weather apps use big numbers and simple icons so you understand the weather in one second.

Level up!

Challenges for weather scientists

Level 1: Comfort meter

Show HOT, NICE or COLD on the screen

Level 2: Sticky alert

Warn when it is hot AND humid

Level 3: Max and min

Remember the highest and lowest temperature

Level 4: Rain chance

Show 'rain likely' when humidity is above 85%

Boss level

Save a reading every 10 minutes and draw a graph of the whole day!

Tip: use **if / else if / else** blocks for the comfort meter. Choose your own numbers for HOT and COLD!

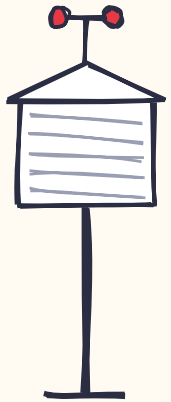
Show what you know

Recap and exit ticket

Exit ticket

- [ ] What does humidity measure, and what is its unit?
- [ ] Which two readings does the DHT22 give us?
- [ ] Why do we wait 2 seconds between readings?
- [ ] Which place was hottest, and why?

At home: check the weather app tonight. Does it match what your station would say?



Future link
In week 14 we put this data on a webpage!

Next week: a Smart Dustbin with a fill-level indicator!